

MVLU COLLEGE

AI PRACTICAL NO. 8

AIM:Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

#CODE:

```
# PRACTICAL 8: ADABOOST ENSEMBLE LEARNING
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)
# Upload and read dataset
files.upload()
df = pd.read_csv("College_Marks_Dataset.csv")
# Prepare data
X = df.drop(columns=["Student_ID", "Name", "Grade"])
y = df["Grade"]

pre = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore"), ["Class"])
], remainder="passthrough")

# Split data
X_train, X_test, y_train, y_test = train_test_split(
```

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```
X, y, test_size=0.20, random_state=42, stratify=y
)

# INDIVIDUAL WEAK CLASSIFIER

weak = Pipeline([
    ("pre", pre),
    ("model", DecisionTreeClassifier(max_depth=1, random_state=42))
])

weak.fit(X_train, y_train)
weak_pred = weak.predict(X_test)
weak_acc = accuracy_score(y_test, weak_pred)

print("\n===== INDIVIDUAL WEAK CLASSIFIER =====")
print("Accuracy:", round(weak_acc * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, weak_pred))
print("\nClassification Report:")
print(classification_report(y_test, weak_pred))

# ADABOOST ENSEMBLE

ada = Pipeline([
    ("pre", pre),
    ("model", AdaBoostClassifier(
        estimator=DecisionTreeClassifier(max_depth=1),
        n_estimators=100,
        learning_rate=0.5,
        random_state=42
    ))
])

ada.fit(X_train, y_train)
ada_pred = ada.predict(X_test)
ada_acc = accuracy_score(y_test, ada_pred)

print("\n===== ADABOOST ENSEMBLE =====")
print("Training completed successfully.")
print("Accuracy:", round(ada_acc * 100, 2), "%")
```

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```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, ada_pred))
print("\nClassification Report:")
print(classification_report(y_test, ada_pred))

# ACTUAL VS PREDICTED

result = pd.DataFrame({
    "Actual Grade": y_test.values,
    "Weak Classifier": weak_pred,
    "AdaBoost": ada_pred
})

print("\n===== ACTUAL VS PREDICTED =====")
print(result.head(15))

# COMPARISON

comparison = pd.DataFrame({
    "Model": ["Weak Classifier", "AdaBoost"],
    "Accuracy": [weak_acc, ada_acc]
})

print("\n===== MODEL COMPARISON =====")
print(comparison)

print(
    "\nAdaBoost Improvement:",
    round((ada_acc - weak_acc) * 100, 2),
    "%"
)

# COMPARISON GRAPH

plt.figure(figsize=(7, 5))
plt.bar(comparison["Model"], comparison["Accuracy"])
plt.xlabel("Model")
plt.ylabel("Accuracy")
plt.title("Weak Classifier vs AdaBoost")
```

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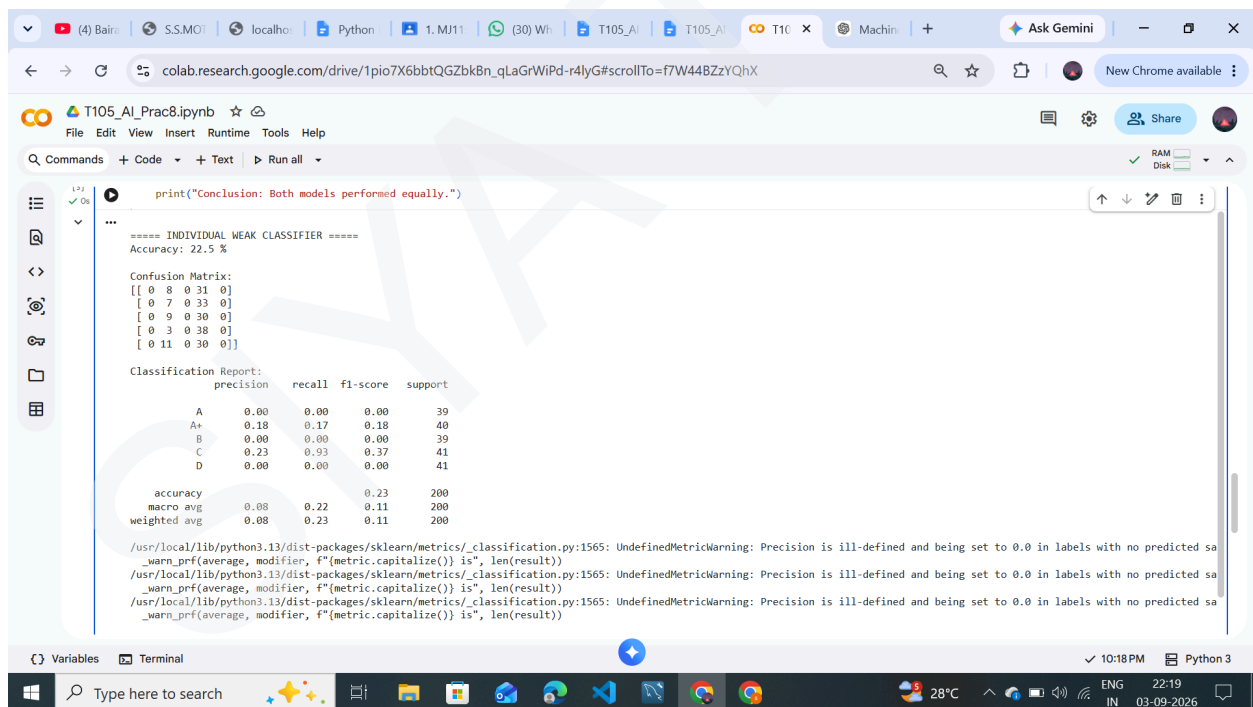
```
plt.ylim(0, 1)
plt.show()
```

```
# FINAL RESULT
```

```
print("\n===== FINAL RESULT =====")
print("Weak Classifier Accuracy:", round(weak_acc * 100, 2), "%")
print("AdaBoost Accuracy:", round(ada_acc * 100, 2), "%")

if ada_acc > weak_acc:
    print("Conclusion: AdaBoost performed better.")
elif ada_acc < weak_acc:
    print("Conclusion: Weak Classifier performed better.")
else:
    print("Conclusion: Both models performed equally.")
```

#OUTPUT:



The screenshot shows a Google Colab notebook interface. The code cell is executed, and the output is displayed. The output includes a confusion matrix and a classification report. The confusion matrix shows the relationship between predicted and actual classes. The classification report shows the precision, recall, f1-score, and support for each class.

=====
Accuracy: 22.5 %

Confusion Matrix:

	A	A+	B	C	D
A	0	8	0	31	0
A+	0	7	0	33	0
B	0	9	0	30	0
C	0	3	0	38	0
D	0	11	0	30	0

Classification Report:

	precision	recall	f1-score	support
A	0.00	0.00	0.00	39
A+	0.18	0.17	0.18	40
B	0.00	0.00	0.00	39
C	0.23	0.93	0.37	41
D	0.00	0.00	0.00	41
accuracy			0.23	200
macro avg	0.08	0.22	0.11	200
weighted avg	0.08	0.23	0.11	200

Warning messages:

```
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted sa
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted sa
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_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

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The screenshot shows a Google Colab notebook titled 'T105_AI_Prac8.ipynb'. The code cell contains the following output:

```
print("Conclusion: Both models performed equally.")
```

==== ADABOOST ENSEMBLE ====

Training completed successfully.
Accuracy: 19.5 %

Confusion Matrix:

```
[[ 2 14  2 14  7]
 [ 5  8  4 10 13]
 [ 4  7  5 13 10]
 [ 1  8  8 14 10]
 [ 4 11  7  9 10]]
```

Classification Report:

	precision	recall	f1-score	support
A	0.12	0.05	0.07	39
A+	0.17	0.20	0.18	40
B	0.19	0.13	0.15	39
C	0.23	0.34	0.28	41
D	0.20	0.24	0.22	41
accuracy			0.20	200
macro avg	0.18	0.19	0.18	200
weighted avg	0.18	0.20	0.18	200

==== ACTUAL VS PREDICTED ====

Actual Grade	Weak Classifier	AdaBoost
0	A	C
1	A	A+
2	C	D
3	B	D
4	A	C
5	A+	C
6	D	C
7	A+	A+
8	B	A
9	B	D
10	A	D
11	A	A
12	D	C
13	A	A+
14	D	B

The screenshot shows a Google Colab notebook titled 'T105_AI_Prac8.ipynb'. The code cell contains the following output:

```
print("Conclusion: Both models performed equally.")
```

==== ACTUAL VS PREDICTED ====

Actual Grade	Weak Classifier	AdaBoost
0	A	C
1	A	A+
2	C	D
3	B	D
4	A	C
5	A+	C
6	D	C
7	A+	A+
8	B	A
9	B	D
10	A	D
11	A	A
12	D	C
13	A	A+
14	D	B

==== MODEL COMPARISON ====

Model	Accuracy
0 Weak Classifier	0.225
1 AdaBoost	0.195

AdaBoost Improvement: -3.0 %

Weak Classifier vs AdaBoost

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Weak Classifier vs AdaBoost

